

Unit 5, Lesson 3: Exploring Angle-Angle-Side and Hypotenuse-Leg Triangle Congruence



Benchmarks

MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



Learning Targets

- ☐ I can use constructions to determine if AAS congruence can be used to show that two triangles are congruent.
- ☐ I can determine if HL congruence can be used to show that two triangles are congruent using deduction.



5.3.1 Warm-Up: Giving Back

Morgan State University is a historically black, public research university in Baltimore, Maryland. In the early 1960s, Calvin E. Tyler Jr., a young African American man who was the first in his family to go to college, attended Morgan State. However, in 1963 he was forced to drop out when he was unable to pay his tuition.



He began working as a truck driver for the United Parcel Service (UPS). Throughout his career at UPS, he continued to be promoted and ultimately served as both the senior vice president for U.S. operations and as a director within the company. On February 23, 2021, he and his wife Tina pledged \$20 million to Morgan State University's endowment to go toward scholarships for students in financial need.

Choose one of the following questions to answer.

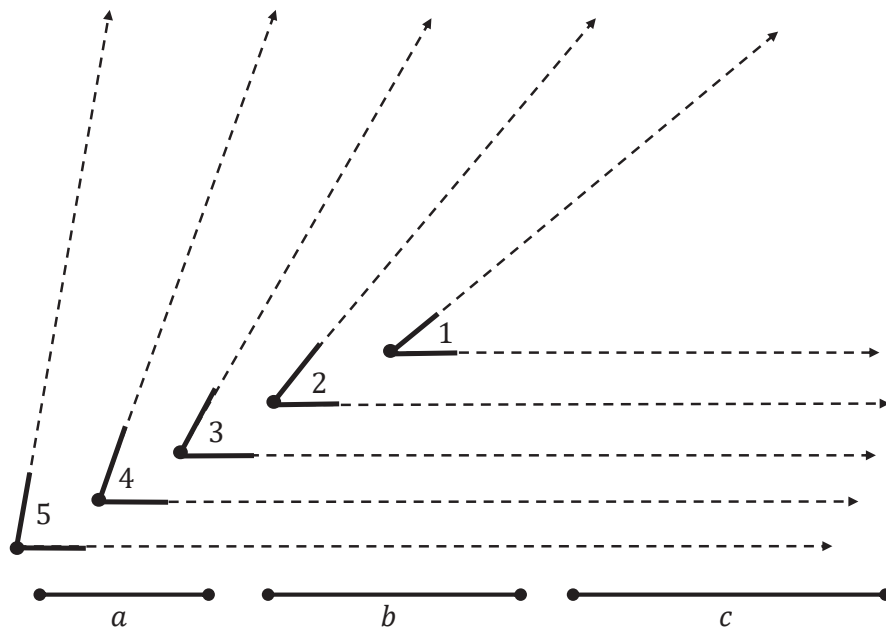
1. How do you think having a strong drive to succeed likely played a role in Mr. Tyler's career?
2. Why do you think it was important to Mr. Tyler to pledge this endowment?





5.3.2 Exploration: Which Other Criteria is Enough?

Five angles labeled 1-5 and three line segments labeled a , b , and c are given.



Use the given parts to create triangles by copying the parts needed onto patty paper using the given criteria.

Helpful Hint: When copying an angle, include some (or all) of the dotted lines so that angles and line segments can be connected.

1. Construct two different triangles using each set of criteria. Use the provided steps in each criteria.

Criteria 1: Given two angles (AA)

Step 1: Copy $\angle 1$ onto a piece of patty paper.

Step 2: Close the triangle using $\angle 2$.

Step 3: Trace the completed triangle. Label $\angle 1$ and $\angle 2$ in the completed triangle.

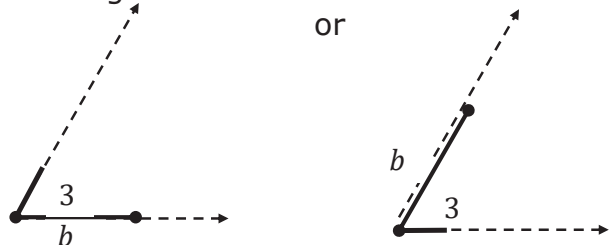
Step 4: Label your triangle "Criteria 1 (AA)."

Step 5: Repeat these steps with $\angle 3$ and $\angle 5$.

Criteria 2:**Given two angles and a non-included side (AAS)**

Step 1: Copy side b onto a piece of patty paper. Label side b on your patty paper.

Step 2: Place $\angle 3$ at one of the ends of side b and trace the ray that does not overlap with side b to copy the angle. Label $\angle 3$.



Step 3: Use $\angle 4$ to close the triangle such that $\angle 4$ is opposite from side b . Label $\angle 4$.

Step 4: Trace the completed triangle. Label your triangle "Criteria 2 (AAS)."

Step 5: Repeat these steps with side c , $\angle 1$, and $\angle 2$.

2. Compare your four triangles with your partner's.

3. Write your partner's name on the line in the table. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	$\angle 1$ and $\angle 2$	$\angle 3$ and $\angle 5$	side b , $\angle 3$, and $\angle 4$	side c , $\angle 1$, and $\angle 2$
My triangle is congruent to _____'s triangle.				

4. With your partner, find another pair of students to form a group of four. Write the two new students' names in the table below. Then, state "true" or "false" to indicate if your triangle is congruent to your partner's triangle for each criteria.

Criteria	AA		AAS	
	$\angle 1$ and $\angle 2$	$\angle 3$ and $\angle 5$	side b , $\angle 3$, and $\angle 4$	side c , $\angle 1$, and $\angle 2$
My triangle is congruent to _____'s triangle.				
My triangle is congruent to _____'s triangle.				

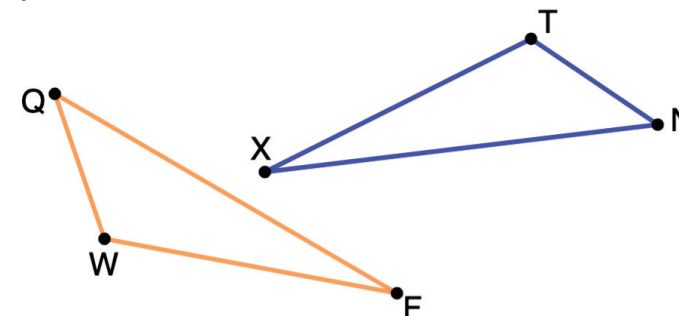
5. Discuss with your group: If given either set of criteria, does that mean all of the parts of the triangles are congruent? Then, write your explanation in the table.

Criteria	Explanation
Given two angles (AA)	
Given two angles and non-included side (AAS)	



5.3.3 Collaboration: Making Connections

1. $\triangle QWF$ and $\triangle NTX$ are shown. $\overline{FQ} \cong \overline{XN}$, $\angle FQW \cong \angle XNT$, and $\angle QWF \cong \angle NTX$.



Two students discussed the information given.

Mabel thinks there is not enough information to determine congruence, because the side given is not between the two angles given.

Xavier thinks the triangles are congruent because he knows that the third angles in the triangles have to be congruent, and then there would be a side between two angles.

With your partner, discuss both students' reasoning. Then, answer the following.

- a. Explain whether you agree with Mabel or Xavier.

- b. What additional information could be added to the argument you agree with to make it more convincing?

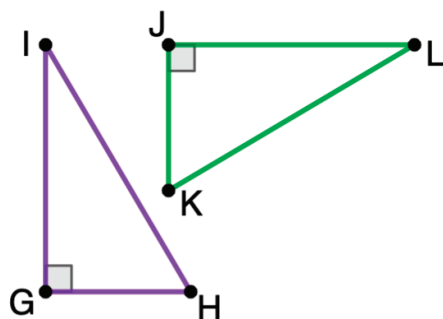
2. Triangle IHG and ΔLKJ are shown. $\overline{IH} \cong \overline{LK}$, $\overline{HG} \cong \overline{KJ}$, and $\angle IGH$ and $\angle LJK$ are right angles.

Eli and Kaylah had the following discussion.

Eli said, "In a situation where you have two sides without the angle in between them, you can't tell if the triangles are congruent."

Kaylah said, "Normally I would agree with you, except these are right triangles."

- a. Discuss with your partner what Kaylah might be thinking.
- b. When given two sides of a right triangle, how can the length of the third side be determined?



Kaylah continued, "For a right triangle, as long as I have two of the sides, I can find the length of the third side. Then I can use SSS since all three corresponding sides would be congruent."

- c. Work with your partner to explain why Kaylah's argument could work to show congruence for any pair of right triangles where the hypotenuse and one leg are identified as congruent.

- d. Explain whether Kaylah's argument would be needed to show congruence between right triangles where both legs of the triangles are identified as congruent.



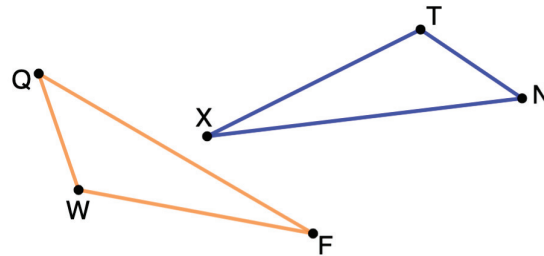
5.3.4 Bring It Together: Triangle Congruence



Angle-Angle-Side (AAS) Congruence Theorem

If two angles and a side opposite one of them are congruent to two angles and the corresponding side of the other, respectively, then the triangles are congruent.

1. Work with your partner to explain how Angle-Angle-Side Congruence Theorem is true. Use the triangles shown to provide an example if desired.



2. Complete the statements.

If two angles of one triangle are congruent to two

angles of another triangle, then the third pair of angles

are ☐ similar.
☐ congruent. If a congruent side opposite one of

the given angles is congruent to the corresponding side

of the other triangle, then ☐ SSS
☐ SAS
☐ ASA also applies. Thus,

AAS is enough to show congruence.



The Hypotenuse-Leg (HL) Theorem

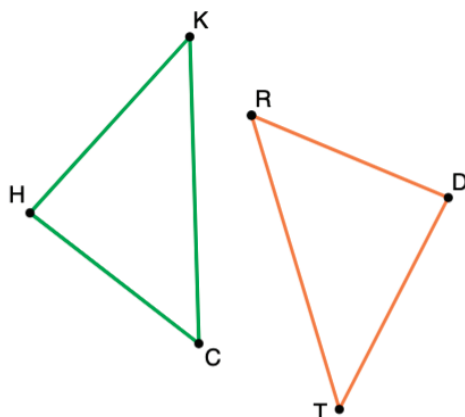
Two right triangles are congruent if the hypotenuse and a leg of one are congruent to the hypotenuse and a leg of the other, respectively.

3. The previous lesson's exploration showed that SSA is not sufficient to prove congruence. Explain why HL is sufficient to prove congruence for right triangles, despite being two sides with a non-included angle.



5.3.5 Try It: Which Criteria?

1. Triangles HKC and DTR are shown. Determine if the given information is sufficient to determine the triangles are congruent. If so, name the congruence theorem or postulate that shows that.



Given Information	Are They Congruent?	Theorem or Postulate Used
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\angle KCH \cong \angle TRD$		
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\angle CHK \cong \angle RDT$		
$\angle KCH \cong \angle TRD$, $\angle HKC \cong \angle DTR$, $\overline{HC} \cong \overline{DR}$		
$\overline{HK} \cong \overline{DT}$, $\overline{KC} \cong \overline{TR}$, $\overline{HK} \perp \overline{HC}$, $\overline{DT} \perp \overline{DR}$		
$\overline{HK} \cong \overline{DT}$, $\overline{HC} \cong \overline{DR}$, $\angle KHC \cong \angle TDR$		



5.3.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each congruence postulate or theorem.



I need help.



I got it.



I could help someone.

Side-Side-Side (SSS)			
Side-Angle-Side (SAS)			
Angle-Side-Angle (ASA)			
Angle-Angle-Side (AAS)			
Hypotenuse-Leg (HL)			